

Per-copy profile HMMs: worked toy example

Addressing your second question: 'how do you USE per-copy profile HMMs to solve the assignment problem?'

How to USE per-copy profile HMMs to assign a read to a copy

Two HMMs: HMM_A (built from Copy A's sequence), HMM_B (built from Copy B's sequence).

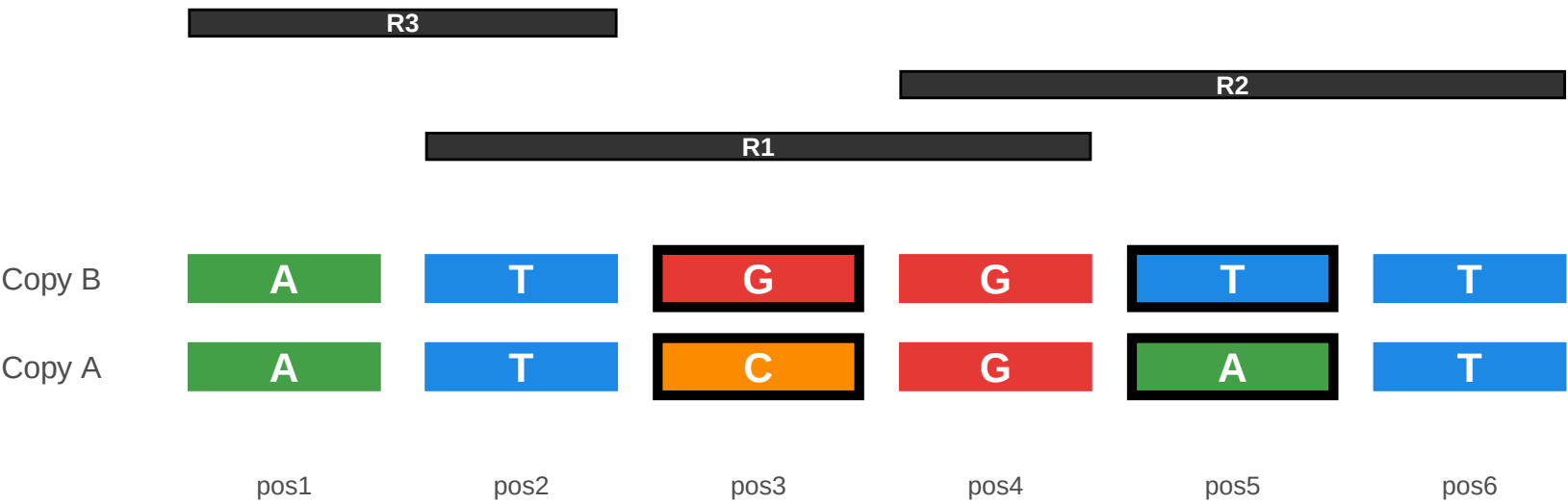
For each match state i : $P(\text{base} = S_c[i] \mid \text{HMM}_c) = 0.95$ and $P(\text{other base}) = 0.017$.

Given a read R with bases $r_1 \dots r_L$ starting at position k :

$$\log L(R \mid \text{HMM}_c) = \sum_i \log P(r_i \mid \text{HMM}_c \text{ at position } k+i-1)$$

$D\log L = \log L_A - \log L_B \rightarrow$ sign of $D\log L$ tells us which copy R came from.

Toy example: copies A, B and 3 reads (R1, R2, R3)



R1: read covers pos 2–4 (includes the SNP at pos 3)

(simulated source: Copy A (read carries A's allele 'C' at pos 3))

read sequence: TCG (covers positions 2–4)

vs HMM_A: $P_{\text{per_pos}} = 0.950 \cdot 0.950 \cdot 0.950$
 $\log L_A = -0.154$

vs HMM_B: $P_{\text{per_pos}} = 0.950 \cdot 0.017 \cdot 0.950$
 $\log L_B = -4.197$

$D\log L = \log L_A - \log L_B = +4.043 \rightarrow$ assign to Copy A

R2: read covers pos 4–6 (includes the SNP at pos 5)

(simulated source: Copy B (read carries B's allele 'T' at pos 5))

read sequence: GTT (covers positions 4–6)

vs HMM_A: $P_{\text{per_pos}} = 0.950 \cdot 0.017 \cdot 0.950$
 $\log L_A = -4.197$

vs HMM_B: $P_{\text{per_pos}} = 0.950 \cdot 0.950 \cdot 0.950$
 $\log L_B = -0.154$

$D\log L = \log L_A - \log L_B = -4.043 \rightarrow$ assign to Copy B

R3: read covers pos 1–2 (no SNP coverage)

(simulated source: ambiguous (covers no diagnostic position))

read sequence: AT (covers positions 1–2)

vs HMM_A: $P_{\text{per_pos}} = 0.950 \cdot 0.950$
 $\log L_A = -0.103$

vs HMM_B: $P_{\text{per_pos}} = 0.950 \cdot 0.950$
 $\log L_B = -0.103$

$D\log L = \log L_A - \log L_B = +0.000$ AMBIGUOUS ($D\log L \sim 0$)

Resulting assignment per read

■ ambiguous ■ Copy A ■ Copy B

